Trainings

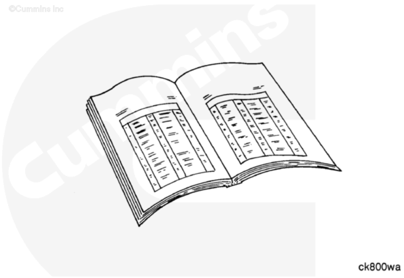
General Information

The backside idler fan belt tensioner system is designed to keep the belt tension stable during transient engine conditions.

The two purposes of the system are to take up belt slack during engine acceleration and restrict idler pulley kick back and belt slack during engine deceleration.

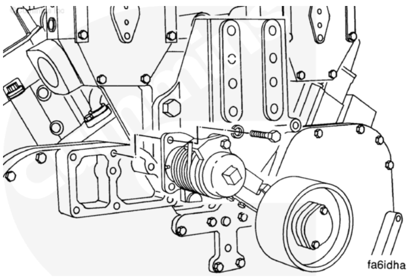
Several types of fan belt tensioning arrangements are used on QST30 engines:

- Shock Absorber
- Linear Tensioner (two types)
- Control rod.



⚠ CAUTION ⚠

Do not use the control rod as a tensioning device. This can cause premature component wear and possible belt malfunction.

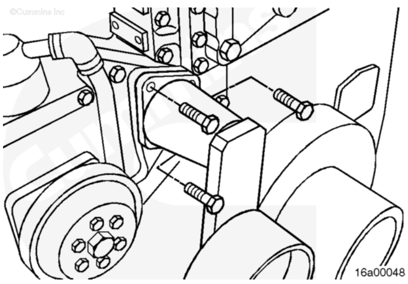


The shock absorber and control rod systems utilize a torsion spring in the fan drive idler arm assembly pivot to take up belt slack during engine acceleration.

The shock absorber and control rod **only** provide resistance during engine deceleration.

Both linear tensioner systems use an enclosed linear spring to regulate belt slack during engine acceleration and deceleration.

Note : If a linear tensioner system is used, the fan belt idler arm does not use a torsion spring. The pivot provides free rotation.



Preparatory Steps

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See the equipment manufacturer service information.
- Remove the cooling fan and spacers. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-040.html)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html)

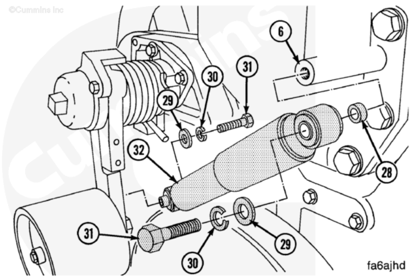
Remove

Shock Absorber System

Make note of washer and spacer location upon removal to verify proper control rod alignment during installation.

Remove the capscrews (31) , washers (29, 30), and spacer (28).

Remove the shock absorber.

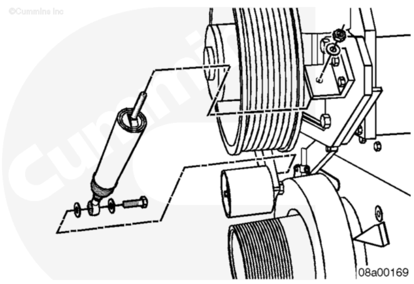


Linear Tensioner System

Remove the capscrews and washers from the lower clevis and idler arm.

Remove the nut and the capscrews from the threaded rod end of the tensioner at the upper anchor bracket.

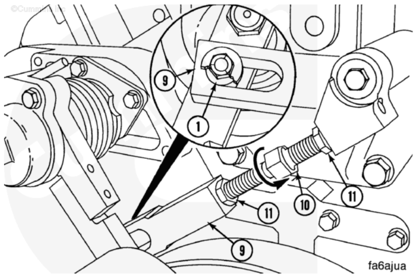
Remove the fan belt tensioner.



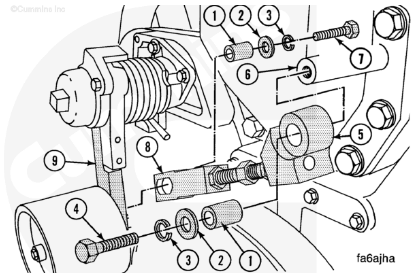
Control Rod System

Note : One of the jam nuts has left-hand threads.

Loosen the jam nuts (11) on the control rod. Turn the adjusting screw (10) until the spacer (1) is **not** touching the end of the slot in the control rod (9).



Remove the capscrews (4) and (7), the washers (2) and (3) and the spacers (1). Remove the control rod assembly.

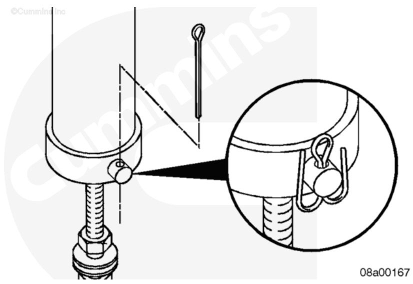


Disassemble

Linear Tensioner System

Remove the cotter pin from the clevis pin.

Remove the clevis pin from the control rod and belt tensioner housing and separate.



Remove the control rod (1) from the control rod retainer (2).

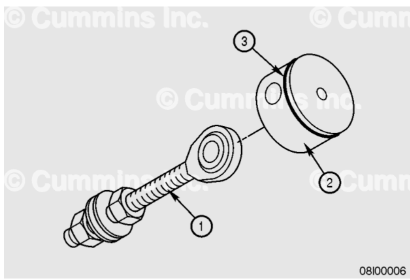
Remove and discard the o-ring (3) from the control rod retainer (2), if present.

Remove the first nut from the control rod threaded end.

Remove the flat washers from the control rod threaded end.

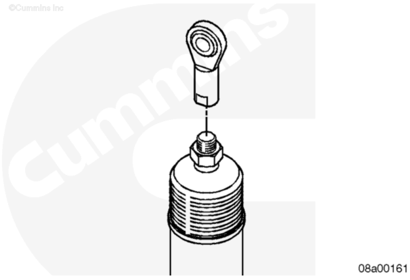
Note : The number of washers may vary. Record the number of washers removed so the same number is reassembled, if required,

Remove the remaining nut from the control rod threaded end.



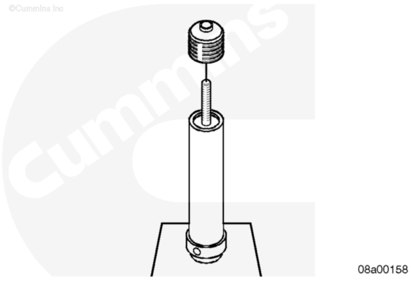
Remove the control rod from the capscrew threaded end.

Remove the nut holding the dust seal to the capscrew.



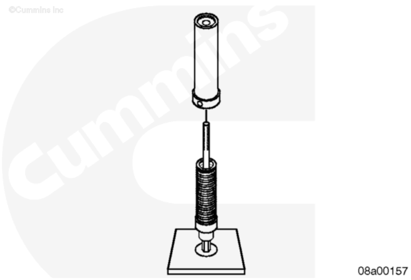
Cut the wire tie to remove the dust seal from the belt tensioner housing.

Remove the dust seal from the threaded end of the capscrew and belt tensioner housing.



Note : It is not necessary to use a build fixture to disassemble or assemble as shown in the illustration. This is an aid to clarify the steps in the procedure.

Remove the belt tensioner housing from the belt tensioner assembled parts.

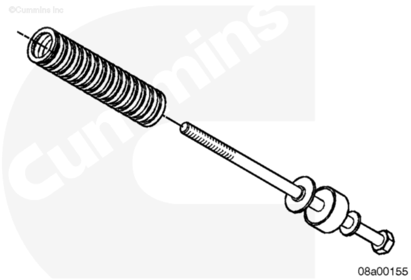


Remove the compression spring from the spring retainer.

Remove the spring retainer from the spring guide.

Remove the spring guide from the flat washer.

Remove the flat washer from the capscrew.

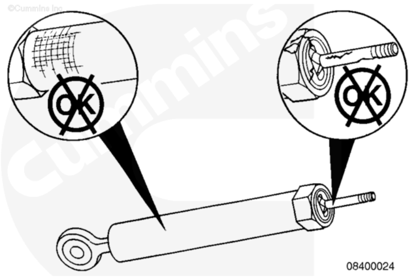


Inspect for Reuse

Linear Tensioner System

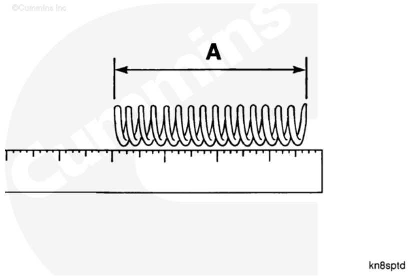
Inspect the tensioner assembly and ends for cracks or excessive wear.

If the tensioner assembly is cracked or worn, it **must** be replaced.



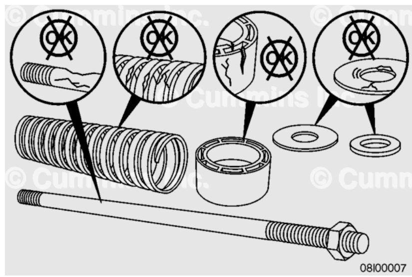
Measure the free length of the spring

Spring Free Length		
mm		in
147.3	MIN	5.80
155.4	MAX	6.12



Note : Replace with new springs if not within specifications.

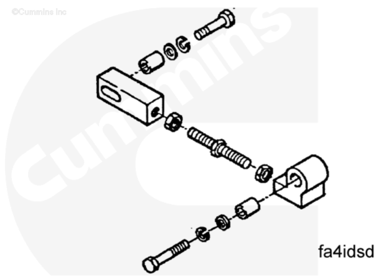
If the tensioner is disassembled, inspect the components for rust or corrosion. All rusted or corroded parts must be replaced.



Control Rod System

Check the parts of the control rod for damage or excessive wear.

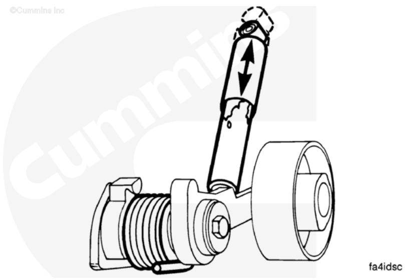
If the assembly is cracked or worn, it **must** be replaced



Shock Absorber System

Check the shock absorber for cracks and fluid leakage.

The shock absorber **must** be replaced if any of the above conditions are found.



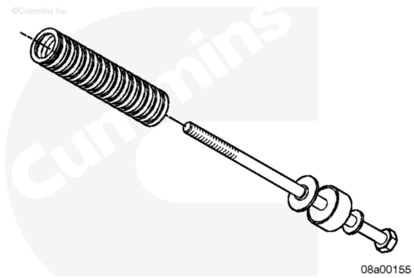
Assemble

Control Rod System

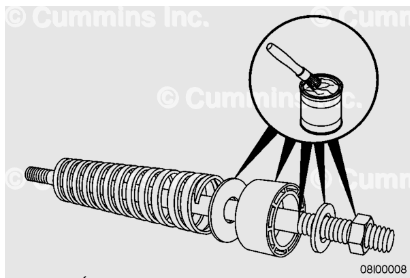
Install a washer on the capscrew so that the round side of the washer faces the capscrew head.

Install the spring guide with the recessed/indented side next to the flat side of the washer.

Install the spring retainer with the larger flat surface against the spring guide.

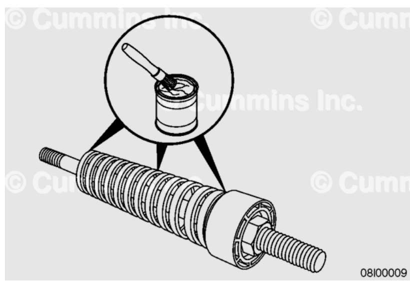


Use Lubriplate™ 105, Part Number 3163087, or equivalent, to grease the spring guide, washers, retainer, and top of the capscrew between the washers, retainer, and spring.

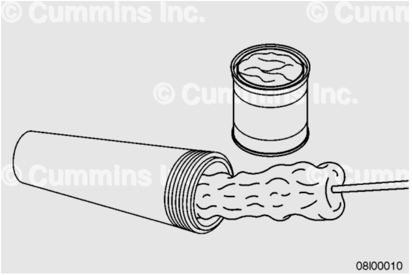


Install the compression spring against the spring retainer.

Use Lubriplate 105, Part Number 3163087, or equivalent, to grease the outside of the compression spring. Make sure to grease both contacting end surfaces.

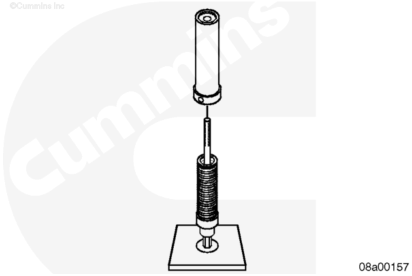


Use Lubriplate 105, Part Number 3163087, or equivalent, to grease the inside of the fan belt tensioner housing.



Note : It is not necessary to use a build fixture to disassemble or assemble as illustrated. This is an aid to clarify the steps in the procedure.

Install the belt tensioner housing over the assembled parts and allow the bottom of the housing to sit on the build fixture with the threads of the capscrews protruding out of the belt tensioner housing.

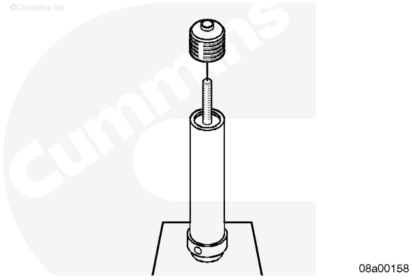


Slip the dust seal over the end of the capscrew and the edge of the belt tensioner housing.

Note : Use a ratchet to prevent the capscrew from moving while pushing the dust seal into place.

Use a wire tie and fasten around the dust seal where it overlaps the belt tensioner housing.

Pull the tie until tight and cut off the excess of the tie.



Push the dust seal down over the threads of the capscrew and install the nut.

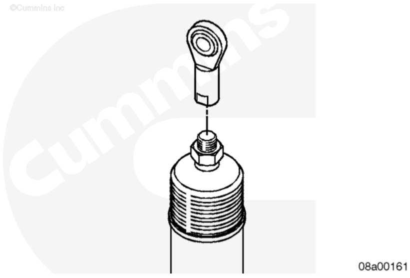
Back the nut against the control rod and tighten.

Use a 3/4-inch open-end wrench to tighten the nut.

Tighten the nut just far enough to get clearance for the following part to be installed. Do **not** completely tighten at this time.

Install the control rod end on the threaded end of the capscrew.

Back the nut against the control rod and tighten.



Torque Value: 100 n•m [74 ft-lb]

Apply Lubriplate™ 105, Part Number 3163087, or equivalent, to the o-ring groove (3) in the control rod retainer (2) and the spring guide.

Install the o-ring (3) on the control rod retainer (2) and install in the housing.

Install the nut on the control rod end (1).

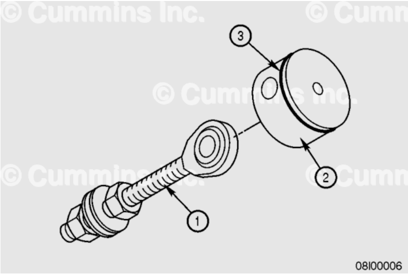
Install flat washers onto the control rod end.

Install the same number of washers that were removed earlier.

Install a second nut onto the control rod end.

Do **not** tighten completely. Leave the washers loose.

Install the assembled control rod end on the control rod retainer on the slotted end.

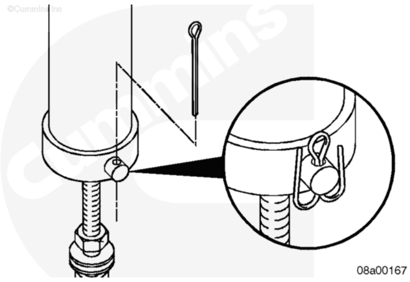


Use a non-metallic mallet, or equivalent, to install the retainer into the belt tensioner housing.

Align the holes and install the clevis pin to hold the parts together.

Use a cotter pin to secure the clevis pin to the housing.

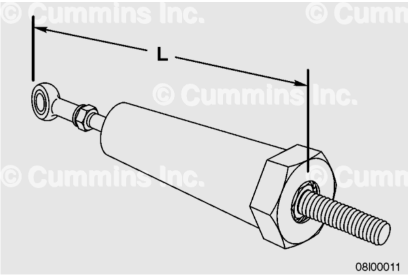
Bend the cotter pin around the clevis pin to keep it from falling out during operation.



Measure the length of the assembly from the top of the cylinder to lower control rod to verify it meets specification.

If the assembly does **not** meet specifications, it **must** be disassembled and assembled or replaced.

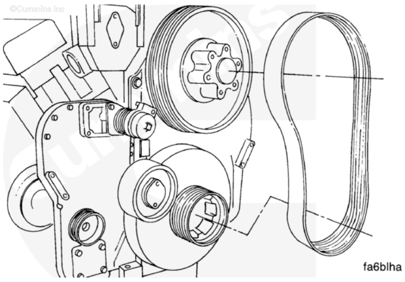
Length of Control Rod Assembly		
mm		in
359	MIN	14.125
362	MAX	14.250



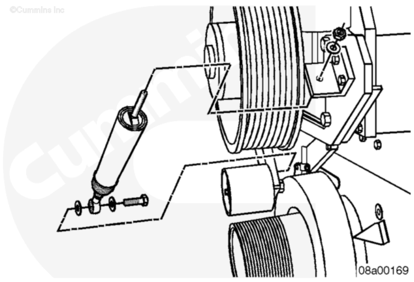
Install

Note : The fan hub pulley and the fan belt are shown removed in the illustrations below for clarity.

Install the fan belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-002-tr.html)



Linear Tensioner System



Attach the lower clevis end of the tensioner to the idler arm with capscrews and washers.

Tighten the capscrews.

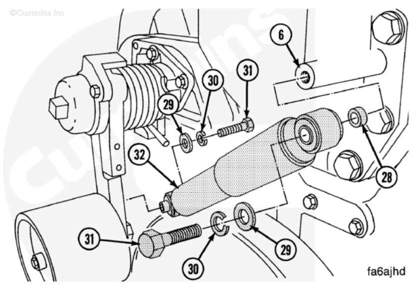
Torque Value: 102 n•m [75 ft-lb]

Slide the threaded-rod end of the tensioner through the hole in the upper anchor bracket and secure it with a washer and nut.

Pull the belt tensioner until the idler pulley contacts the fan belt. Tighten the top nut hand-tight.

Note : Do not tighten the nut. It will be tightened later to adjust the belt tension.

Shock Absorber System



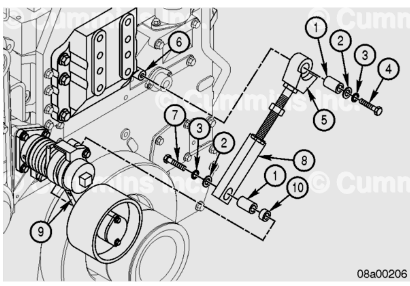
The shock absorber (32) **must** be installed with the larger outer tube of the absorber attached to the fan hub support (6). If the absorber is installed incorrectly, dirt can enter the tube and cause the part to malfunction.

Install the fan hub support end of the shock absorber using the upper capscrew (31), spacer (28), and washers (29, 30).

Install the idler pulley end of the shock absorber using the lower capscrew (31) and washers (29, 30).

Torque Value: 108 n•m [80 ft-lb]

Control Rod System



Install the capscrew (4) in the upper control rod end (5) with spacer (1), washer (2), and lock washer (3) to the fan hub support (6).

Install the capscrew (7) in the lower control rod end (8) with the spacers (1), washer (2), and lock washer (3). Install the lower control rod end through the counterweight washer (10) and into the idler arm (9). Tighten capscrews (4 and 7).

Torque Value: 102 n•m [75 ft-lb]

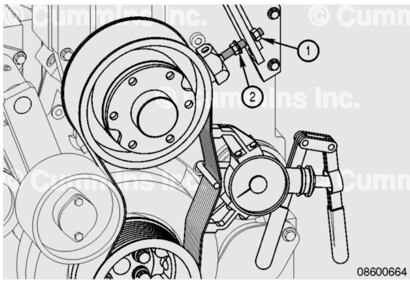
Adjust

Linear Tensioner System

Loosen the jam nut. Place the belt tension gauge, Part Number 3164750, or equivalent, in the middle of the longest belt span between the two pulleys.

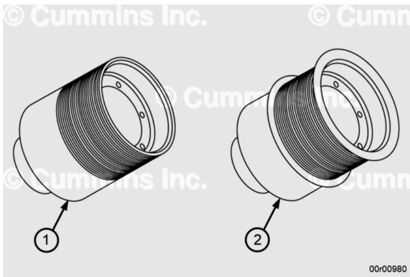
Measure the belt tension.

Tighten the top adjusting nut (1) until the fan belt tension gauge reads between:



Linear Tensioner System			
	n		lb
	2335	MIN	525
	2780	MAX	625

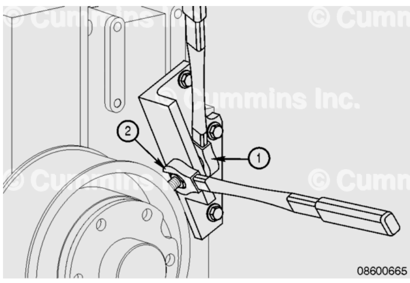
Engines Built with Double Lip Assembly (2)		
n.m		ft-lb
2335	MIN	525
2780	MAX	625



Bar the engine 360 degrees and check the belt tension again. If the tension is no longer within specification, repeat the steps above to adjust the belt tension back into the specified range and bar the engine 360 degrees again. Repeat as necessary until the tension remains in range.

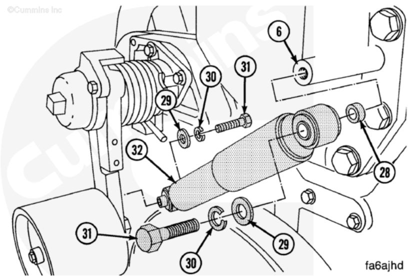
Hold the adjusting nut (1) and tighten the jam nut (2) to specification.

Torque Value: 81 n•m [60 ft-lb]



Shock Absorber System

There is no adjustment required for engines equipped with a shock absorber



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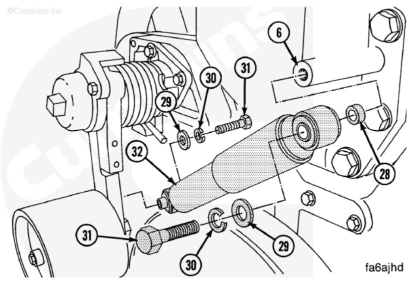
Control Rod System

There is no adjustment required for engines equipped with a shock absorber



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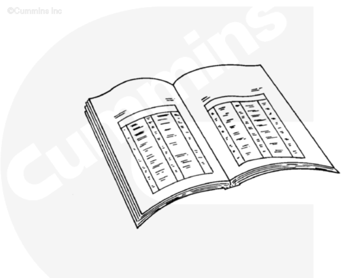
Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the cooling fan and spacers. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-040.html)
- Connect the batteries. See the equipment manufacturer service information.
- If a new fan belt was installed, operate the engine for 10 minutes at high idle to allow for belt stretch.
- Adjust the fan belt tensioner again according to the procedure above in order to take up the slack caused by the initial belt stretch.

Last Modified: 26-Jul-2024